Determinants of Insurance Premiums: Lifestyle Choices vs. Demographic Factors

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1. Introduction

Within 2023, healthcare spending in the United States increased by 7.5%, totaling to 4.9 trillion USD; this healthcare spending increase parallels increases in the cost of health insurance. Recent rates of change in health insurance premium cost has exceeded that of wages and inflation, making it harder for individuals to afford health insurance. However, health insurance costs vary from one individual to another; factors such as age, location and tobacco use are legally allowed to increase the cost of one's premium.

However, individuals with other pre-existing health risks may themselves select costlier healthcare plans that provide greater coverage. Further, other variables may affect tobacco use. Overall, we can categorize factors that potentially drive healthcare cost into two categories: modifiable lifestyle choices and baseline demographic/socioeconomic factors.

Many studies have investigated the role of these variables individually. We try to fill a research gap by investigating how much premium prices can be attributed by baseline demographic/socioeconomic factors (DRF) versus modifiable lifestyle factors (MLC); does one category have a more substantial effect on insurance cost variance? The results indicate how institutions and individuals can make healthcare more affordable. For instance, if insurance cost variance is heavily driven by MLC, customers can lower prices by modifying behaviors. However, if insurance cost is heavily driven by DRF, policy changes may be required to lower costs.

We consider three variables representing MLC: smoking status (current, former, never), alcohol consumption frequency (never, occasionally, weekly, daily), and BMI. We consider six variables representing DRF, age, sex, highest education level, income, marital status, and household size.

Dataset: We utilize the Medical Insurance Cost Prediction dataset from Kaggle [1]. The dataset contains approximately 100,000 observations including demographic details (Age, Sex, Income) and lifestyle indicators (Smoking status, BMI, Alcohol frequency). Constituent data was derived from three main sources: Centers for Disease Control and Prevention (CDC), U.S. Department of Health & Human Services (HHS), and Centers for Medicare & Medicaid Services (CMS).

Research Question: **How much can premium prices be attributed by baseline demographic/socioeconomic factors (DRF) versus modifiable lifestyle factors (MLC)?**

2. Exploratory Data Analysis

2.1 Summary Statistics

Table 1 summarizes the central tendencies of the continuous variables. The substantial difference between the median (463.58 USD) and mean (582.32 USD) premium indicates a right-skewed distribution, suggesting the presence of high-cost outliers.

Table 1: Summary of Numerical Variables

Average Age	Average BMI	Median Premium	Mean Premium	Max Premium
47.52	26.99	463.58	582.32	10962.55

2.2 Distribution and Transformation

Linear regression assumes approximately normal residuals with constant variance. When the response variable is heavily right-skewed, these assumptions are typically violated. As shown in Figure 1 (Left), the raw annual premium distribution exhibits substantial right-skewness. We therefore applied a log-transformation, ln(Premium), which produces a more symmetric distribution (Figure 1, Right). This transformation typically improves residual normality and homoscedasticity, which we verify in Section 4.1.

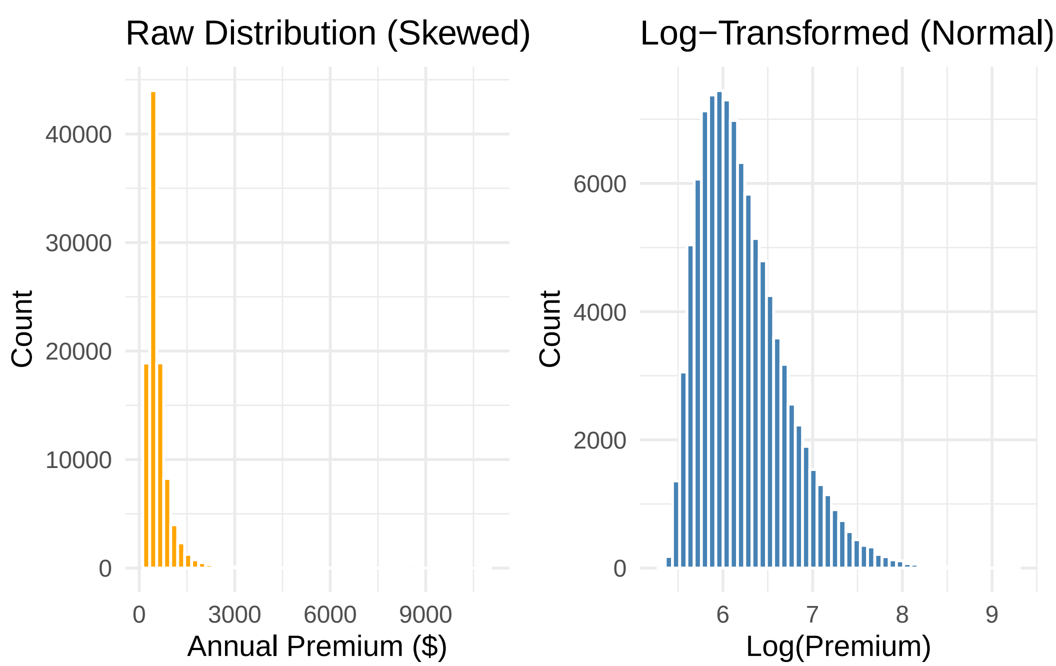

Figure 1: Distribution of annual premiums before and after log-transformation

2.3 Visualizing the “Smoking Penalty” and Interaction

We hypothesized that the cost burden of smoking might vary by age. Figure 2 plots Age against Log-Premium, stratified by smoking status.

The Green Line (Current Smokers) shows a higher intercept, indicating a higher baseline cost. More importantly, the slope appears steeper than that of the Red Line (Never Smokers). This suggests a potential interaction effect: the financial penalty for smoking compounds as the individual ages. We will test this formally in our regression models.

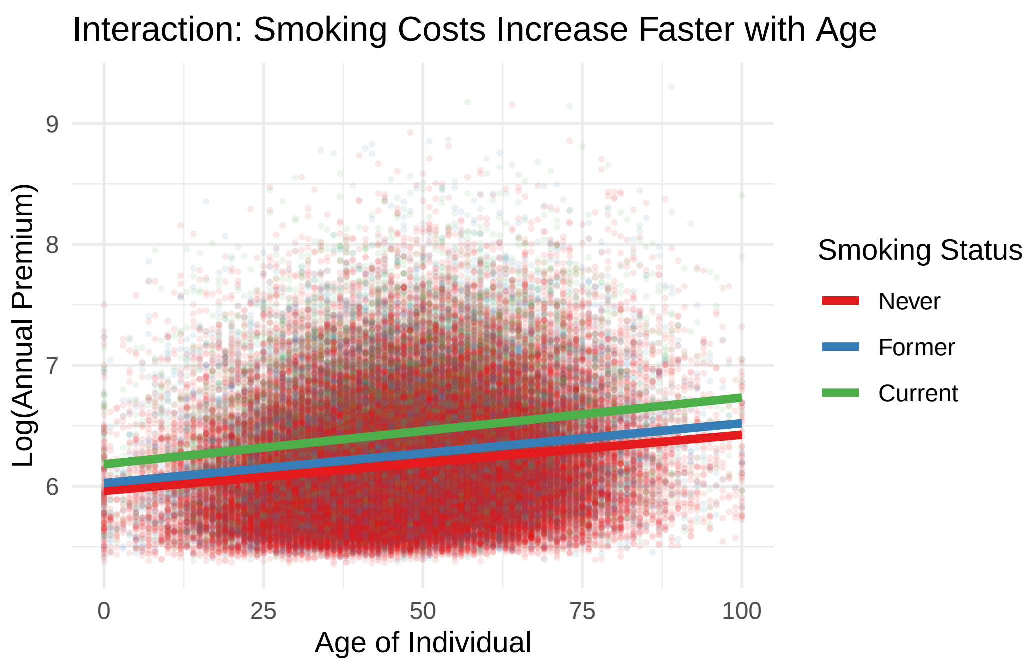

Figure 2: Log-premium by age, stratified by smoking status. The steeper slope for current smokers suggests an interaction effect.

2.4 BMI Trends

We also examined the relationship between BMI and cost. Figure 3 shows a positive (though weak) linear relationship between BMI and the log of insurance premiums. This still allows to consider BMI as a relevant predictor to include in the “Modifiable Lifestyle” category.

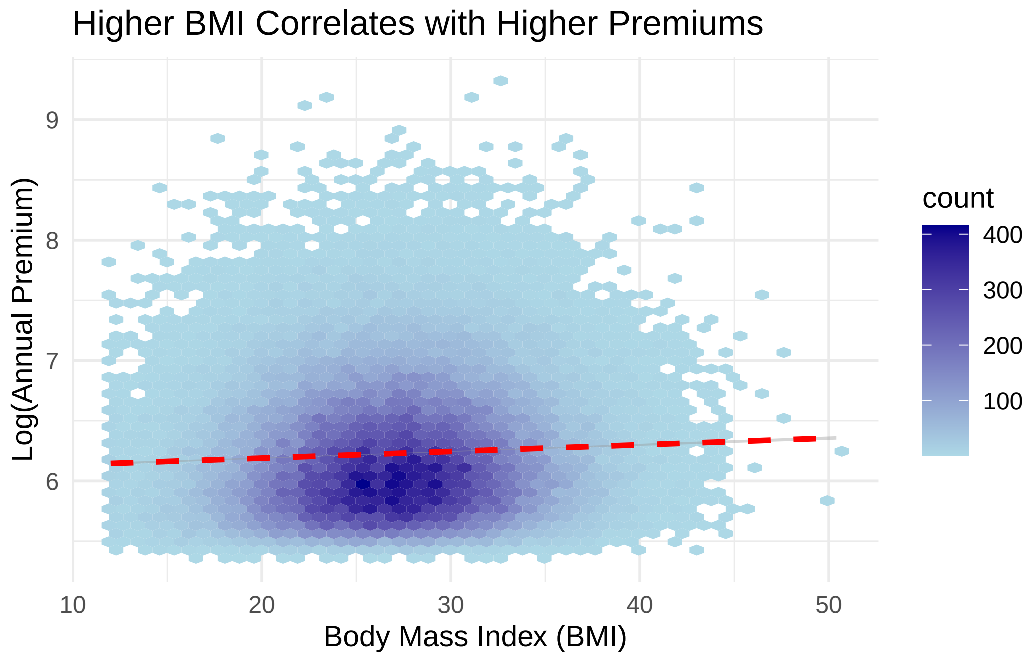

Figure 3: Relationship between BMI and log-transformed annual premium, showing a weak positive association.

3. Methodology

To isolate the variance explained by lifestyle choices, we employ Hierarchical Multiple Linear Regression[3]. We fit three distinct models:

Model 1: Baseline Demographics This model accounts for fixed factors the individual cannot easily change.

$$\ln(Y) = \beta_0 + \beta_{Age} + \beta_{Sex} + \beta_{Region} + \beta_{Income} + \beta_{MaritalStatus} + \beta_{Urban/Rural} + \beta_{Education} + \epsilon$$

Model 2: Lifestyle This model accounts for only modifiable behaviors that the individual can easily change

$$\ln(Y) = \beta_0 + \beta_{Smoker} + \beta_{Alcohol} + \beta_{BMI} + \epsilon$$

Model 3: Full Model (Demographics + Lifestyle) This model adds modifiable behaviors. Based on the interaction observed in Section 2.3, we include an interaction term for Age and Smoking.

$$\ln(Y) = \beta_0 + \beta_{Age} + \beta_{Sex} + \beta_{Region} + \beta_{Income} + \beta_{MaritalStatus} + \beta_{Urban/Rural} + \beta_{Education}$$

$$+\beta_{BMI} + \beta_{Smoker} + \beta_{Alcohol} + (\beta_{Age} \times \beta_{Smoker}) + \epsilon$$

We utilize Analysis of Variance (ANOVA) to statistically compare models 1 and 2 with the Full Model. The F-test in the ANOVA will determine if the reduction in the Residual Sum of Squares (RSS) provided by the lifestyle/demographic variables is statistically significant.

4. Results

4.1 Diagnostics and Assumptions Checking

To ensure the validity of our inference, we examined the residual plots for the full Model.

- Linearity: Figure 4 (Left)
- Equal Variance: Figure 4 (Left)
- Independence:
- Normality of Residuals: Figure 4 (Right) shows the Normal Q-Q plot. The points (circles) adhere closely to the diagonal line for the majority of observations. However, the tails deviate upward. This "heavy-tailed" behavior indicates that our model slightly underestimates the premiums for the most extreme, high-cost outliers. Given the large sample size ($N \approx 100,000$), the Central Limit Theorem ensures that our coefficient estimates remain unbiased despite this deviation.

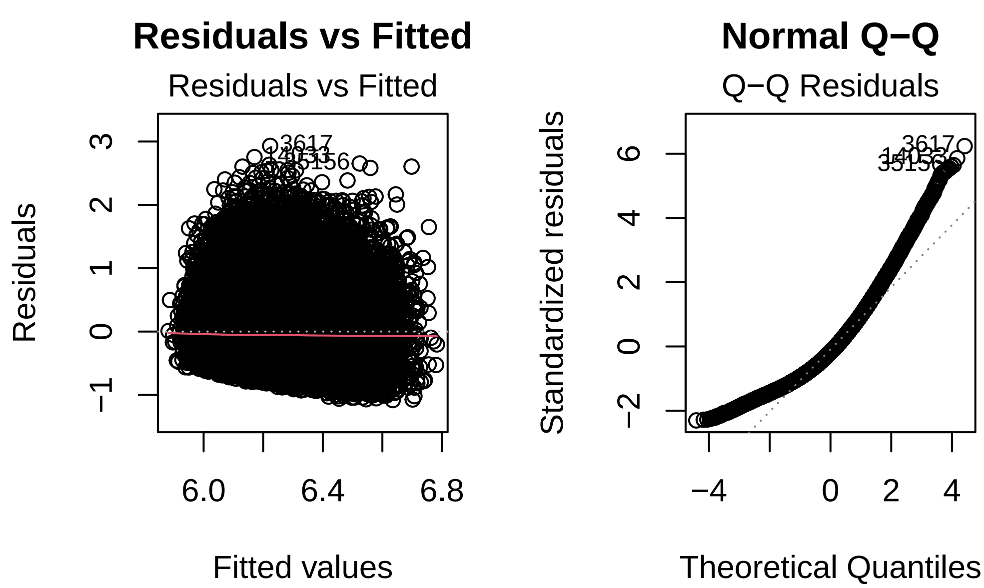

Figure 4

4.2 ANOVA Model Comparison

We performed an ANOVA F-test to compare the explanatory power of Model 1 (Demographics) versus Model 3 (full) and Model 2 (Lifestyle) versus Model 3 (Full).

Table 2: ANOVA Comparison: Demographics vs. Full Model

Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
99980	22900.88	NA	NA	NA	NA
99972	22085.80	8	815.09	461.19	0

Table 3: ANOVA Comparison: Lifestyle vs. Full Model

Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
99993	22683.72	NA	NA	NA	NA
99972	22085.80	21	597.92	128.88	0

Interpretation: The ANOVA table shows a highly significant result for Model 3

Comparing Demographics to Full Model H(null): The additional coefficients (slope) are all zero; therefore, additional variables (lifestlye choices) do not contribute to model fit H(alternative): The additional coefficients (slope) are not all zero; therefore, additional variables (lifestlye choices) do contribute to model fit - F-Statistic: The large F-value (461.19) implies that the variance explained by the added lifestyle variables is significantly greater than what we would expect by random chance - P-Value (< 0.001): Since $p < 0.05$, we reject the null hypothesis. We can conclude that adding Smoking, BMI, and Alcohol significantly improves the model fit over demographics alone. Note: The NA in the first row is normal, as it represents the baseline model against which the second model is compared.

Comparing Lifestyle Choices to Full Model H(null): The additional coefficients (slope) are all zero; therefore, additional variables (demographics) do not contribute to model fit H(alternative): The additional coefficients (slope) are not all zero; therefore, additional variables (demographics) do contribute to model fit - F-Statistic: The large F-value (139.44) implies that the variance explained by the added demographic variables is significantly greater than what we would expect by random chance - P-Value (< 0.001): Since $p < 0.05$, we reject the null hypothesis. We can conclude that adding Smoking, BMI, and Alcohol significantly improves the model fit over demographics alone. Note: The NA in the first row is normal, as it represents the baseline model against which the second model is compared.

As df values are not the same, the f value cannot be used to directly compare the impact of the added demographic variables.

4.3 Comparison of R^2 Values

Table 4: R^2 Values of Demographic, Lifestyle, and Full Model

Category	R2_Value
R^2 Demographic	0.0249303
R^2 Lifestyle	0.0341766
Full Model	0.0596348

Table 5: ΔR^2 of Demographic and Lifestyle Model Compared to Full Model

Category	R2_Value
ΔR^2 (Full Model - Demographic)	0.0347046
ΔR^2 (Full Model - Lifestyle)	0.0254582

ΔR^2

4.4 Regression Coefficients

Table 3 presents the coefficients for the Full Model. Since the response is log-transformed, coefficients (β) can be interpreted as an approximate $100 \times \beta\%$ change in the premium for small values.

Table 6: Coefficients for Modifiable Lifestyle Factors (Model 2)

Predictor	Estimate	Std. Error	P-Value
BMI	0.00562	0.0002976	<0.001
Smoker: Former	0.06373	0.0123031	<0.001
Smoker: Current	0.22166	0.0143493	<0.001
Alcohol: Occasional	-0.00062	0.0034994	0.859
Alcohol: Weekly	-0.00388	0.0042996	0.367
Alcohol: Daily	0.01245	0.0071756	0.083
age:Smoker: Former	0.00032	0.0002462	0.193
age:Smoker: Current	0.00087	0.0002870	0.002

Key Findings:

1. Smoking: Being a "Current Smoker" has a large positive coefficient (0.221), indicating a substantial premium increase compared to never-smokers, holding other factors constant.
2. BMI: The coefficient for BMI is positive and statistically significant ($p < 0.001$). Higher body mass index is reliably associated with higher insurance costs.
3. Alcohol: The p-values for alcohol frequency are > 0.05. This suggests that, unlike smoking and BMI, self-reported alcohol frequency is not a statistically significant predictor of insurance premiums in this dataset.
4. Interaction (Age $\times$ Smoker): The significant interaction term ($p = 0.002$) confirms our EDA finding. The slope of the cost curve is steeper for smokers. This means the dollar-cost of smoking increases as the individual ages.

5. Discussion

6. References

[1] Kaggle. (2024). Medical Insurance Cost Prediction Dataset. [2] Wooldridge, J. M. (2015). Introductory Econometrics. [3] UVa Library. (2023). Hierarchical Linear Regression. https://library.virginia.edu/data/articles/hierarchical-linear-regression [4] American Medical Association. (2024). National Health Expenditure Data. https://www.ama-assn.org/ [5] Bureau of Labor Statistics. (2023). Consumer Expenditure Survey: Health Insurance. [6] HealthCare.gov. (2024). How Insurance Companies Set Health Premiums. https://www.healthcare.gov/how-plans-set-your-premiums/ [7] Kaiser Family Foundation. (2023). Employer Health Benefits Survey. https://www.kff.org/